

QATCH openQCM Real-Time SW Change Control Document

Release Version	Date	Author	Description
v2.7b1	05/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Initial beta version for the v2.7x branch - Updated build information and updater client - New run status ui - Drying mode indicator - Multiprocessing improvements - Always display 10 points in analyze - Add CSV export with selectable columns - Multiple unhandled runtime exceptions during analyze - Version 2.0 VisQ.AI UI and Backend - Restore fill line filtering and tighten point threshold - Visqai performance optimizations - Adjust heating PID parameters for performance - Add new YOLO model assets - Erroneous difference curve inversion triggering - Head tail spike clipping, fix thread degradataion - Modify fill classifier status messages - Simplify/standardize dry detection messaging - Warning indicators for long fills - Move dry detection tolerances to constants.py - Add db_sync module for app.db - Refine dry detection and extend fill warning workflows - Add fill complete message to primary plot - Add 3min initial fill timeout - Add mutable buffer pH to Run Info for VisQ.AI - Manifest driven multi-module loading for .visq files - Automatic cartridge lock with manual override - Resolving CBD encryption bug and downstream issues - Missing average viscosity and stdev in output files - Deprecate embedded progress bars in run and analyze - Suppress redundant session expiration prompts on idle - Missing _stopFreq attribute error when peak way off - Fix peak out of frequency range in serial - Fix possible IndexError in Analyze.run() - Refactor _update_plot() and upgrade performance - Various Analyze Module Improvements - Bug fixes for DryingDetection and FillClassifier stalls - Add Recover tab to Export UI with placeholder msg - Fix drop effect under-correction and reduce noise jitter - Concentration-based surface tension correct formula - Introduce new fill method for visco and shear calulation - Embed progress bars for initialize, qmodel, and analyze - Optimize load run sequence in Analyze - Restrict analysis points to 60% only by removing 80% - Resolve y-axis label and text overlap issues in analysis

SW Changes

v2.7r1	05/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Same change log as v2.7b1
v2.7b2	05/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Initial beta version for the v2.7x branch - Updated build information - Recover DB if corrupted on application start-up - Fix updater so r-builds can switch from v2.6x to v2.7x
v2.7r2	05/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Same change log as v2.7b2
v2.7b3	05/19/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Proposed fix for Live Fill Detection thread issue - Show "Stopped" when user clicks "Stop" button - Fix label overflow where long text lines are clipped
v2.7r3	05/20/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Proposed fix for Live Fill Detection thread issue - Show "Stopped" when user clicks "Stop" button - Fix label overflow where long text lines are clipped
v2.7b4	05/22/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - rate limiting to 0.05s across the board - fix rolling behavior for fast and long runs - labels in status overlay unreadable on smaller screens - Refactor 'RingBuffer' and Patching FillClassifier/DryDetection Bugs

SW Changes

v2.7r4	05/22/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - rate limiting to 0.05s across the board - fix rolling behavior for fast and long runs - labels in status overlay unreadable on smaller screens - Refactor `RingBuffer` and Patching FillClassifier/DryDetection Bugs
v2.7b5	07/09/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Drop effect improvements - Adding QModel Onyx - Fix #419: Initial Fill Trendline Check Uses Single Point - Prediction-thread leaks and crashes in VisQ.AI mode - Fix memory leaks in Analyze.py to boost performance - Fix Temperature Control Buffer Bug - Adding tools UI with Integrated Online Calculators
v2.7b6	07/15/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - QModel Onyx bug running volta refinement detector - Increase QModel dry detection window - Adding QModel stability condition to live fill - Help menu checkmark sync issue fixed
v2.7r6	07/15/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Drop effect improvements - Adding QModel Onyx - Fix #419: Initial Fill Trendline Check Uses Single Point - Prediction-thread leaks and crashes in VisQ.AI mode - Fix memory leaks in Analyze.py to boost performance - Fix Temperature Control Buffer Bug - Adding tools UI with Integrated Online Calculators - QModel Onyx bug running volta refinement detector - Increase QModel dry detection window - Adding QModel stability condition to live fill - Help menu checkmark sync issue fixed
v2.7b7	07/21/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix #411: Drop effect among other r-build bugs

SW Changes

v2.7r7	07/21/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix #411: Drop effect among other r-build bugs
v2.7b8	09/08/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Upgrading QModel Onyx assets to 7.1.1 - Drop effect correction improvements - Fix #434: Bundle calculator resources with app - Fix #436: Fill-line only data produces invalid results - Fix #442: Fill-line analysis data source changed
v2.7r8	09/08/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Upgrading QModel Onyx assets to 7.1.1 - Drop effect correction improvements - Fix #434: Bundle calculator resources with app - Fix #436: Fill-line only data produces invalid results - Fix #442: Fill-line analysis data source changed
v2.7b9	09/09/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix issue with FW update in frozen SW builds
v2.7r9	09/09/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix issue with FW update in frozen SW builds
v2.7b10	10/02/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Improved security checks - Improved surface tension calculation - Improved Export to CSV Report functionality - BUG FIX: Recall Run Info From XML error - BUG FIX: Off-by-one index on partial fills - BUG FIX: Various minor Analyze improvements

SW Changes

v2.7b11	10/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Update to Resources v53 - Improved surface tension calculation - Improved dry detection constants
v2.7r11	10/06/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Update to Resources v53 - Improved security checks - Improved Export to CSV Report functionality - Improved surface tension calculation - Improved dry detection constants - BUG FIX: Recall Run Info From XML error - BUG FIX: Off-by-one index on partial fills - BUG FIX: Various minor Analyze improvements
v2.7b12	10/07/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix #450: Bioformulation Viscosity Corrections - Fix #450: Initial Fill Only Analysis Issues - Fix #450: Done Status Persistence Issues
v2.7r12	10/08/2026	A. Ross	SWEEP PERFORMANCE: FW: 1.5-7.5ms E2E: 1.5-7.5ms CAL: <2s - Updated build information - Fix #450: Bioformulation Viscosity Corrections - Fix #450: Initial Fill Only Analysis Issues - Fix #450: Done Status Persistence Issues